

BETTLES, AK, USA

WMO: 701740

Lat: 66.917N

Lon: 151.515W

Elev: 193

StdP: 99.03

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26533

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.2	(c) -42.3	(d) -37.5	(e) 0.1	(f) -33.9	(g) -36.3	(h) 0.1	(i) -33.1	(j) 8.1	(k) -11.9	(l) 7.1	(m) -11.5	(n) 0.3	(o) 30	(p) 0.721

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	26.0	15.3	24.1	14.5	22.4	13.8	16.6	23.4	15.7	21.8	14.8	20.4	2.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	10.3	17.5	13.2	9.7	17.0	12.2	9.1	16.4	47.1	23.4	44.4	21.8	41.9	20.2	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.7	5.6	5.0		-46.9	28.9	2.7	1.6	-48.9	30.1	-50.5	31.0	-52.0	32.0	-54.0	33.1
			DB	-38.1	18.3	3.5	1.4	-40.6	19.3	-42.7	20.1	-44.6	20.9	-47.2	22.0
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.2	-23.5	-19.0	-15.7	-4.0	7.4	14.6	15.4	11.5	5.5	-5.7	-17.4	-21.1
	DBStd	15.73	11.22	8.83	7.37	6.66	5.35	3.38	3.18	3.47	4.38	6.43	8.27	8.81
	HDD10.0	5622	1038	811	796	420	114	4	3	22	143	486	821	963
	HDD18.3	8259	1297	1044	1054	670	341	117	101	212	386	744	1071	1221
	CDD10.0	421	0	0	0	0	32	143	169	70	7	0	0	0
	CDD18.3	17	0	0	0	0	1	6	9	1	0	0	0	0
	CDH23.3	246	0	0	0	0	11	93	121	21	0	0	0	0
	CDH26.7	31	0	0	0	0	1	11	18	2	0	0	0	0
Wind	WSAvg	2.2	1.8	2.1	2.3	2.7	2.6	2.3	2.2	2.0	2.1	2.1	2.0	1.9
Precipitation	PrecAvg	364	21	18	17	16	19	36	53	61	48	29	25	23
	PrecMax	632	87	82	91	78	76	91	138	150	122	97	98	78
	PrecMin	175	0	0	0	0	3	2	8	14	3	3	2	3
	PrecStd	101	21	17	19	16	15	19	30	30	32	18	23	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.1	0.8	3.7	12.6	24.8	28.1	28.8	26.0	18.6	9.6	0.9	-0.7
		MCWB	-1.6	-0.2	0.2	5.8	12.9	15.6	16.3	16.9	10.5	6.4	-0.9	-1.5
	2%	DB	-2.6	-1.6	0.7	9.7	21.6	25.8	26.4	22.8	16.5	5.8	-1.6	-3.7
		MCWB	-3.3	-2.8	-1.3	4.2	11.4	14.4	15.8	15.0	9.9	3.2	-2.5	-4.5
	5%	DB	-5.3	-4.0	-1.8	7.6	18.9	24.0	24.5	20.2	14.5	3.9	-3.7	-6.1
		MCWB	-5.9	-4.8	-3.8	2.8	10.0	13.6	15.3	13.6	9.1	2.0	-4.3	-6.7
	10%	DB	-8.1	-6.7	-4.4	5.6	16.6	22.3	22.5	17.9	12.3	2.0	-6.4	-9.0
		MCWB	-8.7	-7.6	-6.1	1.5	9.1	13.1	14.7	12.5	8.3	0.5	-7.0	-9.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.6	-0.1	0.9	6.2	13.3	17.0	17.9	17.9	11.9	5.7	-0.6	-1.3
		MCDB	0.0	0.5	2.9	12.0	22.2	24.6	24.7	23.8	16.9	7.3	0.4	-0.5
	2%	WB	-3.0	-2.6	-1.1	4.5	12.1	15.7	16.9	15.8	10.6	3.8	-2.3	-4.3
		MCDB	-2.2	-1.6	0.6	8.9	19.8	22.8	23.8	21.1	14.6	5.4	-1.6	-3.7
	5%	WB	-5.2	-4.7	-3.6	3.1	10.8	14.8	16.1	14.5	9.8	2.1	-4.2	-6.5
		MCDB	-4.6	-3.8	-1.8	7.1	17.8	21.8	22.6	18.6	13.5	3.5	-3.5	-6.0
	10%	WB	-7.7	-7.2	-6.1	1.8	9.6	13.8	15.3	13.4	8.8	0.7	-6.9	-9.3
		MCDB	-7.2	-6.5	-4.4	5.2	15.9	20.7	21.1	16.9	12.0	1.8	-6.4	-8.8
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.8	13.2	11.6	11.0	11.5	10.5	10.0	8.7	7.0	7.1	7.7
		MCDBR	8.2	9.4	12.7	11.2	14.3	14.6	14.5	13.5	12.7	7.9	7.2	7.7
		MCWBR	7.7	8.4	10.5	6.9	6.6	6.1	6.1	6.9	7.3	5.6	6.4	7.2
	5% WB	MCDBR	8.2	9.3	12.0	10.3	12.8	12.9	12.2	11.0	10.5	6.8	6.9	7.3
		MCWBR	7.6	8.5	10.1	6.5	6.2	5.8	5.6	6.0	6.3	4.9	6.3	6.9
Clear-Sky Solar Irradiance	taub	0.168	0.232	0.256	0.310	0.351	0.361	0.367	0.357	0.313	0.265	0.170	N/A	
	taud	1.919	2.154	2.250	2.177	2.296	2.374	2.383	2.426	2.492	2.228	1.934	N/A	
	Ebn at Noon	223	627	815	842	843	844	824	793	753	562	196	0	
	Edh at Noon	13	48	75	108	108	103	99	86	63	44	11	0	
All-Sky Solar Radiation	RadAvg	0.08	0.66	2.46	4.64	5.48	5.68	4.76	3.51	2.09	0.78	0.16	0.00	
	RadStd	0.01	0.06	0.15	0.26	0.49	0.44	0.35	0.36	0.28	0.12	0.02	0.00	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.84	N/A	N/A	N/A	N/A	N/A	N/A	-274	N/A
(47) Regional (3 neighbors)		+0.96	N/A	N/A	N/A	N/A	N/A	N/A	-335	N/A

Nomenclature: See separate page